DIPLOMA PROGRAMME PRACTICE · HL · SET A

Chemistry / Paper 1A + 1B

Worked mark scheme · 75 marks · 120 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer A The 36-neutron isotope has mass number 65. With heavy fraction x, 63(1−x)+65x=64.30, giving x=0.650.

PAPER 1A · QUESTION 2 [1]

Answer A The probabilities are 0.75^2, $2\times0.75\times0.25$ and 0.25^2. Two arrangements produce the mixed-isotope ion.

PAPER 1A · QUESTION 3 [1]

Answer A The large jump after the third removal indicates exhaustion of the valence shell. The fourth electron is appreciably closer to the nucleus and less shielded.

PAPER 1A · QUESTION 4 [1]

Answer A $\lambda=c/\nu$ gives $\lambda_{long}/\lambda_{short}=2.40/1.60=1.50$. $E=h\nu$ is larger at higher frequency.

PAPER 1A · QUESTION 5 [1]

Answer D The sample contains 0.0400 mol C and 0.0800 mol H. Their masses are 0.480 and 0.080 g. The remaining 0.320 g is 0.0200 mol O, giving 2:4:1.

PAPER 1A · QUESTION 6 [1]

Answer D $p_2=p_1(V_1/V_2)(T_2/T_1)=96.0\times1.50\times0.900=129.6$ kPa.

PAPER 1A · QUESTION 7 [1]

Answer B Vaporization separates intact HF molecules. Strong hydrogen bonding accounts for the unusually large intermolecular attraction.

PAPER 1A · QUESTION 8 [1]

Answer B FC=valence−nonbonding−half bonding electrons. The three values are 6−2−3, 6−4−2 and 6−6−1.

PAPER 1A · QUESTION 9 [1]

Answer C ClF_3 has five domains with two equatorial lone pairs. BrF_5 has six domains with one lone pair.

PAPER 1A · QUESTION 10 [1]

Answer D Three C–H bonds and three heavy-atom connections give six sigma bonds. The double bond contributes one pi bond and the triple bond two.

PAPER 1A · QUESTION 11 [1]

Answer A Delocalized electrons carry current in both solids. Differently sized atoms distort the lattice and make slip harder.

PAPER 1A · QUESTION 12 [1]

Answer C Twelve initially separate molecules require eleven connections to make one open chain. Each ester-forming condensation releases one H_2O.

PAPER 1A · QUESTION 13 [1]

Answer D For an isoelectronic series, the electron shells are comparable. Al^{3+} has 13 protons and attracts the electron cloud most strongly.

PAPER 1A · QUESTION 14 [1]

Answer B The absorbed energy is hc/λ. The ratio $\Delta_{480}/\Delta_{600}$ is 600/480=1.25.

PAPER 1A · QUESTION 15 [1]

Answer A Both have formula C_3H_8O, but the first is an alcohol and the second an ether.

PAPER 1A · QUESTION 16 [1]

Answer C The oxygen-bound methyl is the 3H singlet. The quartet is a CH_2 next to carbonyl carbon and adjacent to CH_3, consistent with methyl propanoate.

PAPER 1A · QUESTION 17 [1]

Answer B Three equivalent methyl groups attached to a carbon with no H give a 9H singlet. The OH and mass fit $(CH_3)_3COH$.

PAPER 1A · QUESTION 18 [1]

Answer A C2 in 2-bromobutane has four different substituents. A racemate has zero net rotation and the other two compounds are achiral.

PAPER 1A · QUESTION 19 [1]

Answer C The solution gains 80.0×4.18×3.00=1003 J. The reaction loses this heat, so $\Delta H = -1.003/0.0100 \approx -100$ kJ mol^{-1}.

PAPER 1A · QUESTION 20 [1]

Answer C Reverse the second step to go from Y to Z. Its enthalpy becomes −17, giving −42−17=−59 kJ mol^{-1}.

PAPER 1A · QUESTION 21 [1]

Answer A Lattice formation completes the cycle, so its enthalpy is −320−410=−730. Dissociation has the opposite sign.

PAPER 1A · QUESTION 22 [1]

Answer D $-1367 = 2(-394)+3(-286)-\Delta Hf°(\text{ethanol})$. The products sum to −1646, so $\Delta Hf° = -279$.

PAPER 1A · QUESTION 23 [1]

Answer C $\Delta G° = 36.0 - 0.120T$ in kJ mol^{-1}. It is negative when T exceeds 300 K; at 300 K it is zero.

PAPER 1A · QUESTION 24 [1]

Answer C K<1 gives positive ΔG°=−RT ln K. Because Q/K=0.25, ΔG=RT ln(Q/K) is negative.

PAPER 1A · QUESTION 25 [1]

Answer C $C_3H_8+5O_2\rightarrow3CO_2+4H_2O$. Oxygen limits the reaction to 0.900/5=0.180 mol propane, producing 0.540 mol CO_2.

PAPER 1A · QUESTION 26 [1]

Answer D A larger exposed surface increases collision frequency. Equal amounts of the limiting carbonate eventually yield equal amounts of gas.

PAPER 1A · QUESTION 27 [1]

Answer B A is second order. After its factor four, tripling B supplies factor three, so B is first order. Divide rate units by concentration cubed.

PAPER 1A · QUESTION 28 [1]

Answer D The slow elementary step gives $k_2[I][B]$. Substituting $[I]=K[A][B]$ gives $k_2K[A][B]^2$.

PAPER 1A · QUESTION 29 [1]

Answer D The slope equals −Ea/R. Ea=5700×8.31=47,367 J mol^{-1}.

PAPER 1A · QUESTION 30 [1]

Answer D Compression favours the side with fewer gas molecules. The equilibrium constant depends on temperature.

PAPER 1A · QUESTION 31 [1]

Answer A Forming 0.600 mol HI consumes 0.300 mol of each reactant, leaving 0.200 mol each. $Kc=0.600^2/0.200^2$.

PAPER 1A · QUESTION 32 [1]

Answer C Each concentration halves. The squared numerator falls fourfold while the denominator falls twofold. Qc<Kc, favouring NO_2.

PAPER 1A · QUESTION 33 [1]

Answer B OH^- converts HA to A^-, leaving 0.0500 and 0.0300 mol. $pH=4.80+\log_{10}(0.0300/0.0500)=4.58$.

PAPER 1A · QUESTION 34 [1]

Answer B $x=10^{-2.50}=0.003162$. $Ka=x^2/(0.100-x)=1.03\times10^{-4}$.

PAPER 1A · QUESTION 35 [1]

Answer D Mg atoms oxidize to ions at the anode. The electrons pass along the wire to the copper cathode where Cu^{2+} is reduced.

PAPER 1A · QUESTION 36 [1]

Answer B E°cell=0.80−(−0.76)=1.56 V. Two moles of electrons are transferred per mole of Zn, so ΔG°=−2FE°.

PAPER 1A · QUESTION 37 [1]

Answer B Spontaneous oxidation of M^{2+} requires a reduction couple above +0.40 V. Only X_2 gives a positive cell potential, +0.25 V.

PAPER 1A · QUESTION 38 [1]

Answer A The first equation propagates the chain by replacing Cl• with an ethyl radical. Homolysis starts the chain and radical recombinations terminate it.

PAPER 1A · QUESTION 39 [1]

Answer B In SN2 the nucleophile attacks as the leaving group departs. Both bonds are partial at the transition state.

PAPER 1A · QUESTION 40 [1]

Answer C The secondary carbocation is more stable than the primary carbocation. Bromide attacks the more commonly formed intermediate.

PAPER 1B · QUESTION 1 [12]

Competing kinetic models

(a) Determine both partial orders and the rate expression at 300 K. [3]

1 Runs 1 and 2: doubling A at fixed B quadruples rate, so order in A is two.

1 Runs 2 and 3: doubling B at fixed A doubles rate, so order in B is one.

1 rate=k[A]²[B].

(b) Calculate k at 300 K, give its units, and predict the initial rate at [A]=0.150 and [B]=0.0800 at 300 K. [3]

1 k=0.000500/(0.100²×0.100)=0.500.

1 For total order three, units are $dm^6\ mol^{-2}\ s^{-1}$.

1 Rate=0.500×0.150²×0.0800=9.00×10^{-4} mol $dm^{-3}\ s^{-1}$.

(c) Use runs 1 and 4 to estimate Ea. Use R=8.31 J K^{-1} mol^{-1} and assume the rate law and Arrhenius prefactor apply over this range. [3]

1 Concentrations match, so $k_{315}/k_{300}=2.00$.

1 Ea=R ln(2)/(1/300−1/315).

1 Ea≈36.3 kJ mol^{-1}.

(d) Assess the claim that the rate law proves a single elementary three-particle collision. Suggest how to reduce bias when extracting initial rates. [3]

1 Overall kinetic orders do not by themselves establish a unique mechanism.

1 A fast pre-equilibrium before a slower elementary step can produce the same concentration dependence.

1 Acquire many early-time points and estimate the initial tangent before significant reactant depletion; accept a justified fitting procedure.

PAPER 1B · QUESTION 2 [12]

Equilibrium by absorbance

(a) Use the calibration to determine the equilibrium complex concentration. Explain why the intercept must be used. [3]

1 A=0.025+5000c; net complex signal is 0.900.

1 c=0.900/5000=1.80×10⁻⁴ mol dm⁻³.

1 Ignoring the positive background assigns excess absorbance to the complex and overestimates its concentration.

(b) Calculate both remaining reactant concentrations and Kc using equilibrium concentrations in mol dm⁻³. [3]

1 1:1 consumption leaves [Fe³⁺]=8.20×10⁻⁴ mol dm⁻³.

1 [SCN⁻]=6.20×10⁻⁴ mol dm⁻³.

1 Kc=[complex]/([Fe³⁺][SCN⁻])≈354 using the stated concentration convention.

(c) Water doubles the volume at the same temperature. Compare the immediate Qc with Kc and predict the subsequent change in complex amount. [3]

1 All three concentrations initially halve.

1 Qc increases by (1/2)/(1/2)²=2, so Qc=2Kc.

1 The reaction shifts left, decreasing the amount of complex from its immediate post-dilution value.

(d) Explain why a lower absorbance immediately after dilution does not, by itself, prove a shift. Design one comparison that isolates the shift. [3]

1 Dilution alone halves the complex concentration before chemical re-equilibration.

1 For pure dilution the predicted absorbance is 0.025+0.900/2=0.475 if background is unchanged.

1 Compare final equilibrated absorbance with 0.475 using matched blanks and the same path length; a lower net signal supports further dissociation.

PAPER 1B · QUESTION 3 [11]

Locating a weak-acid endpoint

(a) Find the original analytical concentration of HA and derive pKa from the half-equivalence reading. [3]

1 The 1:1 endpoint corresponds to $0.1000\times0.02000=0.002000$ mol HA.

1 $c=0.002000/0.02500=0.08000$ mol dm^{-3}.

1 At 10.00 cm^3 equal amounts HA and A^- give $pH=pKa=4.80$.

(b) Using the conjugate pair ratio, predict the pH at 15.00 cm^3 and compare with the reading. [3]

1 OH^- forms 1.500 mmol A^- and leaves 0.500 mmol HA.

1 $pH=pKa+\log_{10}(3)=5.28$.

1 The prediction agrees within the displayed precision, supporting the buffer model in this region.

(c) Indicator M changes at pH 3.1–4.4 and N at 8.2–10.0. Choose an indicator and explain the basic equivalence pH. [2]

1 Choose N because its transition lies within the steep section.

1 The conjugate base A^- reacts with water: $A^-+H_2O\rightleftharpoons HA+OH^-$.

(d) At 10.00 cm^3 the buffer is diluted twofold with water. Predict both its approximate pH and its buffer capacity per unit volume. State a limit to the pH prediction. [3]

1 The HA:A^- ratio is unchanged, so pH remains about 4.80.

1 Both buffer concentrations halve; capacity per unit volume decreases.

1 At very large dilution, ionization of water and changes in dissociation can invalidate the simple concentration-ratio approximation.